BV

HACKATHON ROUND 3 • FINAL PROJECT REPORT

BRANDVAULT AI

Protect Every Word. Preserve Every Identity.

The final product:
three promised improvements delivered, measured and live.

3 / 3	223	0.825	99.5%
Round 3 improvements delivered	automated tests, passing on SQLite and PostgreSQL	evidence recall@3 (was 0.675)	OCR accuracy on held-out creatives

LIVE PRODUCT

brandvault-ai-1.onrender.com
API: brandvault-ai.onrender.com

SOURCE CODE

github.com/roshanvikram/BRANDVAULT-AI

TEAM DEAL DYNAMOS

Deepak Mouriya M — Team Leader, Product & Frontend
Roshan Vikram AV — AI / Backend / Full-Stack
Pritheesh Kumar V — MCP, Data & RAG
Aishwarya R M — Visual DNA & QA

REPORT DATE

9 November 2026

OPERATING PRINCIPLE

Agents advise — humans decide.

Supersedes: Round 2 Project Report (7 November 2026)

All figures are measured on the final code

CONTENTS

What this report covers

The Round 3 submission items in the order requested: the final project, the completed improvements, technical details, testing notes, changes to the plan, and team contributions.

01	Summary & links	Working product, repository, how to try it
02	Completed improvements	The three Round 3 promises, with evidence
03	Round 2 bugs fixed	Reliability of the core workflow
04	Technical details	Architecture, modules, design decisions
05	Testing notes	223 tests, three measured evaluations
06	Changes to the plan	What differs from the Round 2 promises, and why
07	Known limitations	Stated openly
08	Demo script	Five-minute walkthrough of the final product
09	Team contributions	Filled in by the team

01 · SUMMARY

A complete, governed workflow — now for images, pipelines and meaning

Round 2 delivered the governance platform and 12 intelligence engines. Round 3 kept every promise from the Round 2 report: BrandVault now reads the text inside creatives, guards publishing from outside tools, and finds evidence by meaning. Several bugs were also fixed that would have broken a real deployment.

1 · Multimodal Creative Inspection

Banners, PDF decks and videos are read with OCR or the exact PDF text, checked like typed copy, and every issue is boxed on the creative. Logo clear space is measured.

2 · Publishing Gate & Integrations

API keys, a Check API, a publishing gate that passes only passport-approved copy, signed webhooks, channel and disclaimer agents, a real MCP server and notifications.

3 · Semantic Evidence Engine

Word, HTML and spreadsheet ingestion, meaning-based search with pgvector on PostgreSQL, version-aware documents and the Guideline Gap Agent.

Try it

Live product	`https://brandvault-ai-1.onrender.com`
API & interactive docs	`https://brandvault-ai.onrender.com/docs`
Source code	`https://github.com/roshanvikram/BRANDVAULT-AI`
Demo sign-in	NOVA TECH workspace accounts (Administrator, Reviewer, Creator) listed in `DEPLOYMENT.md`

> The free host sleeps after 15 idle minutes; the first visit can take up to a minute and the app shows a “waking up the server” banner meanwhile.

02 · COMPLETED IMPROVEMENTS

1 Multimodal Creative Inspection

Upload a banner, social image, PDF deck or short video. BrandVault reads the text, checks it against Brand DNA, Product Truth and ClaimDNA, and draws each issue on the creative.

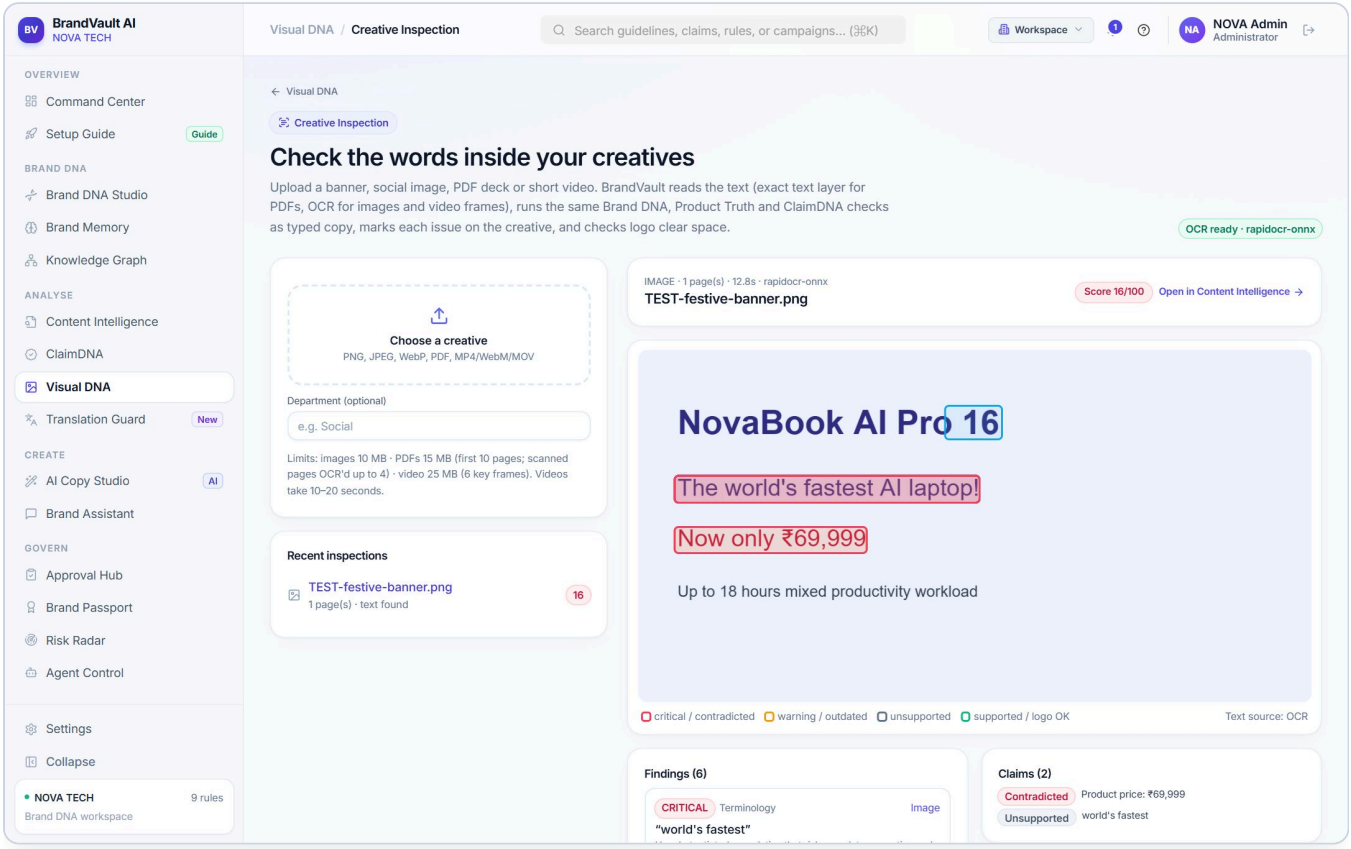


Figure 2.1 — Creative Inspection. A banner uploaded through the real API. The superlative and the ₹69,999 price (approved: ₹79,999) are boxed where they appear; the OCR read "AI" and dropped the ₹, and both were corrected against the brand vocabulary and listed for the reviewer.

PROMISED	DELIVERED
OCR with boxes for images and PDFs	RapidOCR (ONNX, CPU, about 170 MB of RAM). PDFs use the exact text layer; scanned pages are OCR'd at higher resolution
Text flows into ClaimDNA, Product Truth, Terminology	Every creative becomes a normal submission with findings and claims
Logo clear space and placement	Logos are located with Asset Forensics, and edge margins and text intrusion are measured against the ratio set in Settings
Region overlays	Every finding and claim is boxed on its page. The most severe box is drawn on top, and clicking a finding highlights it
Video and deck support	Up to 10 PDF pages; 6 key frames per video
25+ labelled creatives with measured accuracy	26-creative set plus a 12-creative held-out set: issue F1 1.00 on both, OCR character accuracy 99.7% / 99.5%

2 Publishing Gate & Channel Integrations

Checks now run where content is written and published, and nothing gets out without a valid Brand Passport.

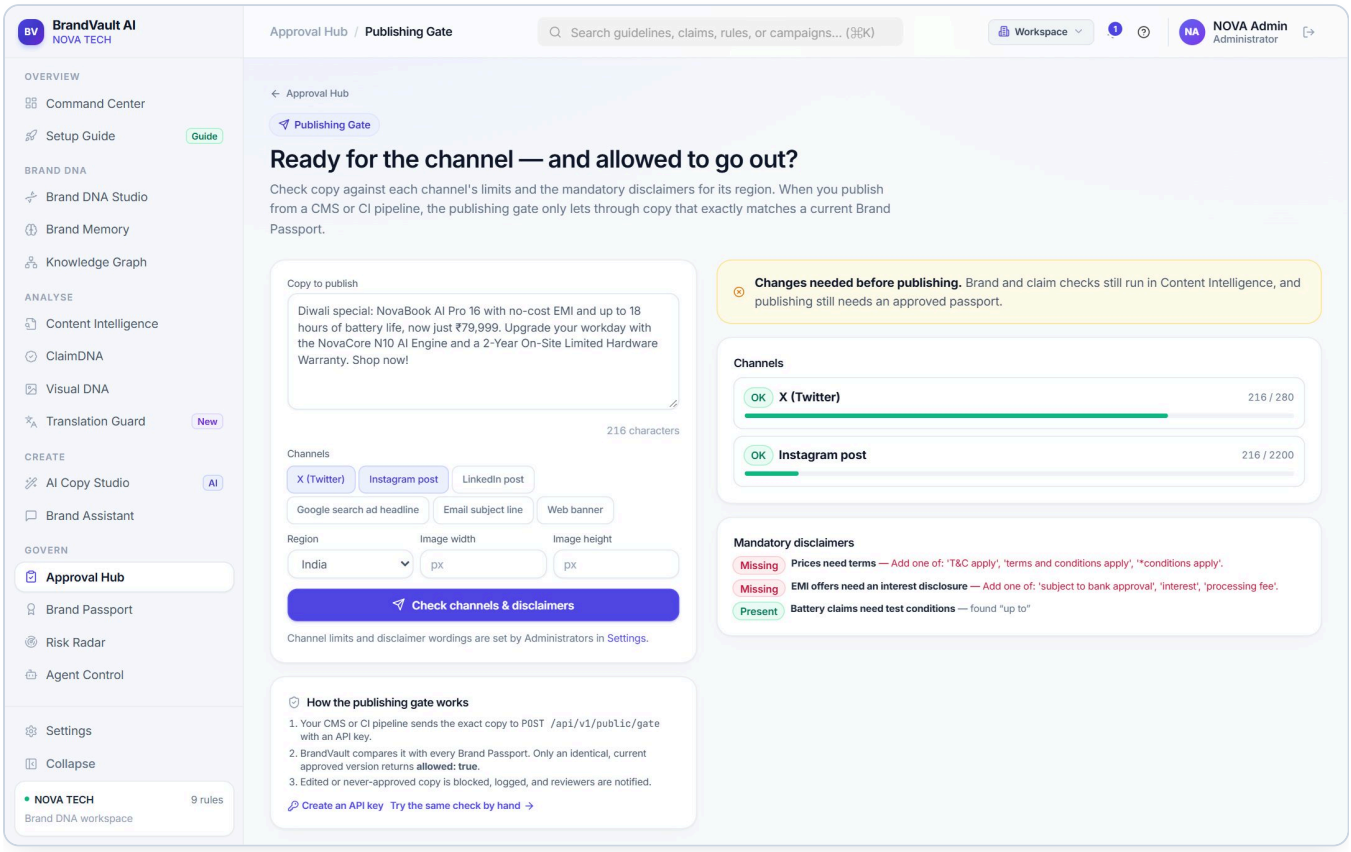


Figure 2.2 — Publishing Gate page. A 216-character post fits X and Instagram. The Disclaimer agent flags the missing price T&C and EMI disclosure and accepts the battery wording (“up to”).

PROMISED	DELIVERED
Public Check API with API keys	Keys are hashed, scoped (check, gate, read), shown once, revocable and usage-counted. Checks can run as dry-run or saved
Signed webhooks	HMAC-SHA256 over timestamp and body, 3 attempts, delivery log and test button; 6 event types
Publishing gate	`POST /public/gate` allows only copy identical to a current passport. Blocked attempts are logged and reviewers are notified
Channel Adapter and Disclaimer agents	Limits, aspect ratios, required elements and regional disclaimers, all editable by Admins in Settings
Full MCP transport	A real MCP server (official SDK 2.x, stdio) with 5 read-only tools, verified with a real MCP client
Notifications	An in-app notification centre for real events, filtered by role. Slack and email are optional, and email shows “not configured” without SMTP
Block 100% of unauthorised calls	Tested: missing key, revoked key, wrong scope and another organisation's data

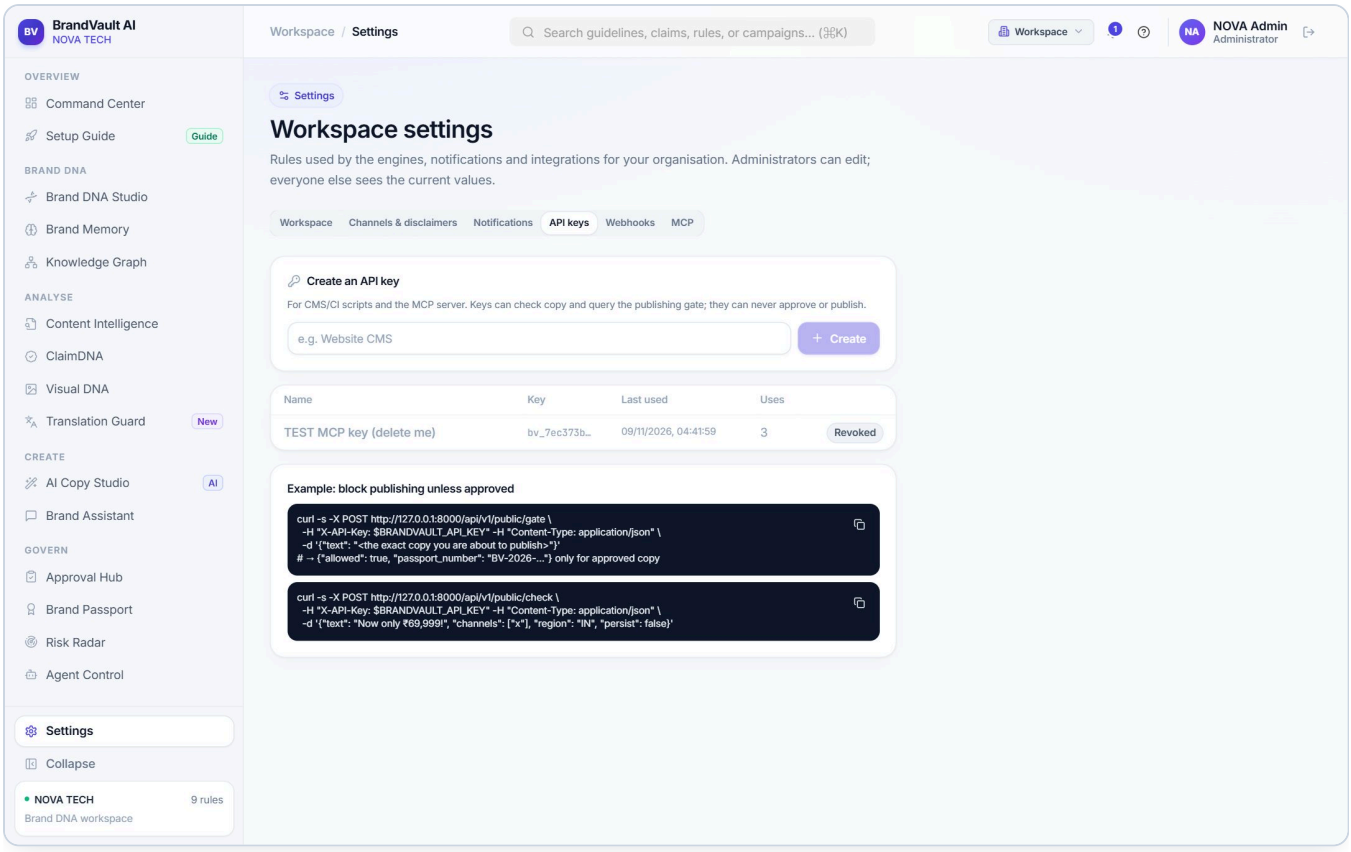


Figure 2.3 — Settings → API keys. Keys are created and revoked by Administrators, with copy-ready examples for a CMS or CI publishing step.

3 Semantic Evidence Engine

PROMISED	DELIVERED
Dense embeddings in pgvector, BM25, RRF	BGE-small embeddings (local, no API key; or any OpenAI-compatible API). BM25 and TF-IDF are fused first, then RRF is applied with the dense ranking. On PostgreSQL a native pgvector table with an HNSW index is used
DOCX, HTML and spreadsheets	PDF, DOCX, HTML, Markdown/text, CSV and XLSX, cited by page, section or row. Passages are sentence-aligned at about 600 characters. Price-list rows become facts awaiting approval
Evidence across document versions	A newer version retires the old document. It stays on record but is no longer cited, and facts taken from it are flagged for re-confirmation
Guideline Gap Agent	Flags product values that disagree across current documents. Sentences that state a value as forbidden are ignored
Better recall, precision kept	Retrieval recall@3 rose from 0.675 to 0.825 and MRR from 0.575 to 0.717. The text analyzer is unchanged at precision 1.00 and recall 0.86

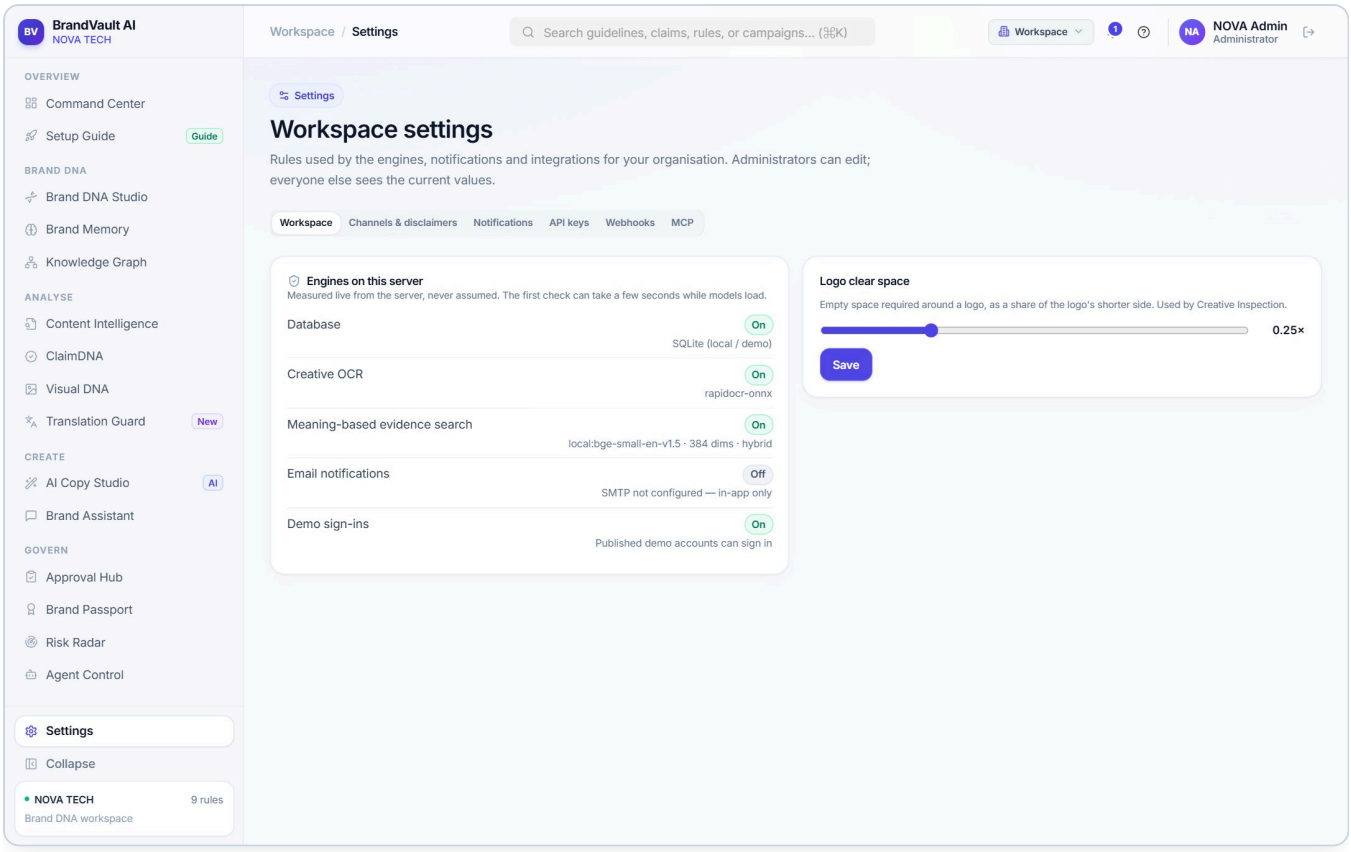


Figure 2.4 — Settings → Workspace. Engine status is measured live from the server: database, OCR, meaning-based search (BGE-small, 384 dimensions, hybrid), email and demo sign-ins.

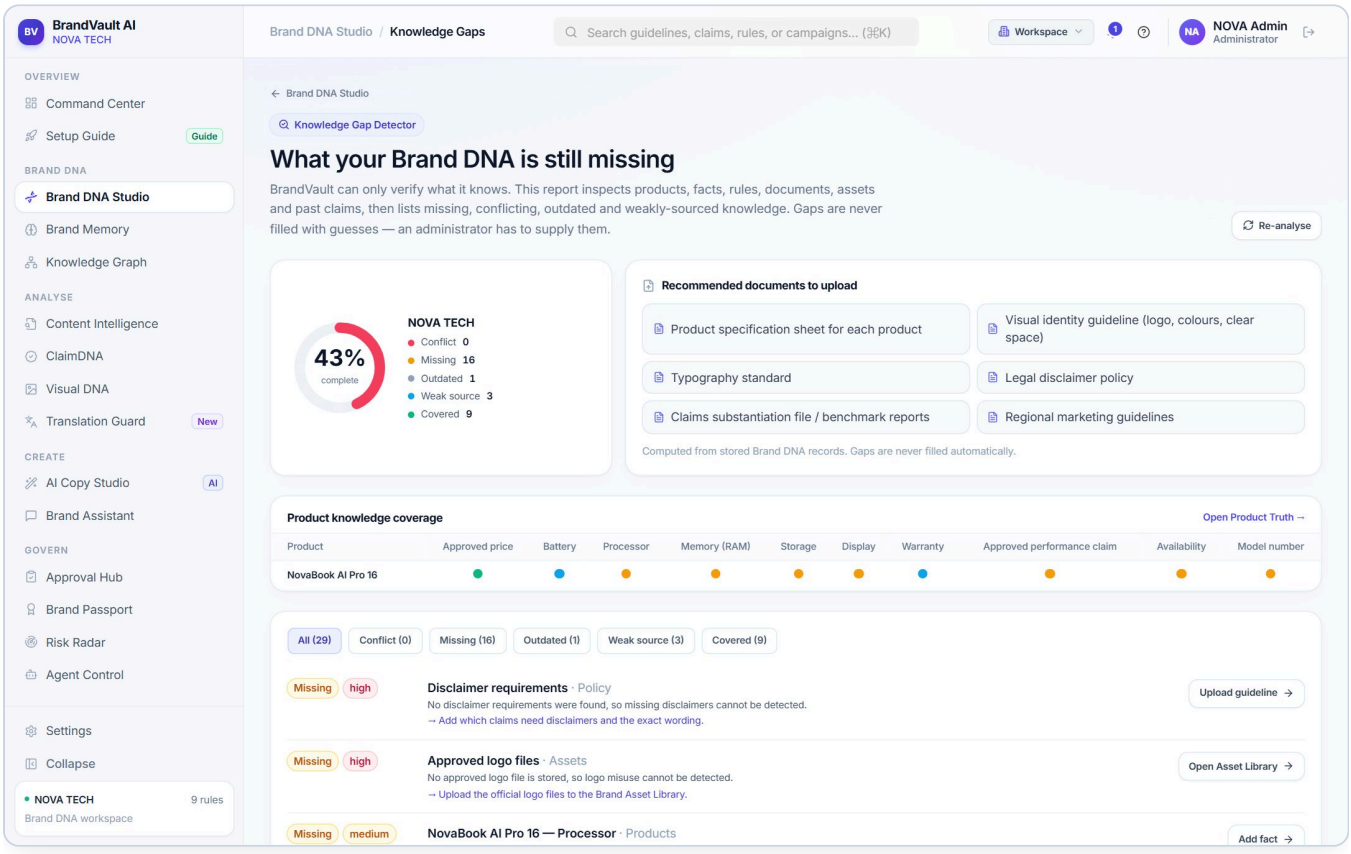


Figure 2.5 — Knowledge Gap Detector. Now includes the Guideline Gap Agent (conflicting values across documents) and facts whose source document was replaced.

03 · ROUND 2 BUGS FIXED

A reliable core workflow

Testing on a real PostgreSQL database and auditing every screen turned up problems that would have broken the product in real use. All are fixed and covered by tests.

PROBLEM	FIX
PostgreSQL would crash at startup: SQLAlchemy 2.1 needs psycopg 3, but requirements shipped psycopg2	Driver changed; the full suite now runs on PostgreSQL 18
Brand Passport failed on PostgreSQL: the guideline version is longer than its 50-character column, which SQLite never enforced	Column widened
Uploads were lost on every Render restart	Files are stored in the database
Migrations worked only on SQLite	Portable additive migrations
Hosted poolers (PgBouncer) break server-side prepared statements	Prepared statements disabled for PostgreSQL
Visual DNA flagged colours against a "monochrome rule" no organisation had, naming NOVA TECH for every brand	Colours are judged only against the organisation's own confirmed palette rule
Visual DNA rejected JPEG and WebP	Supported
A whole PDF page became one passage, so citations were coarse	Sentence-aligned passages
Guideline upload errors were invisible in the UI	Error shown
A malformed .docx caused a server error	Clear 422 message
Settings showed "CONNECTED" even when the health check failed	Rebuilt; shows measured status
The notification bell was guessed from submissions	Real events
The old MCP script read the database directly and defaulted to the NOVA organisation without any key	Replaced with an API-key-scoped, read-only MCP server
Cold start on the free host looked like a broken site	Wake-up banner
A new deployment opened with an empty dashboard	Demo campaigns are seeded from real analyzer output; demo sign-ins can be switched off

04 · TECHNICAL DETAILS

Architecture of the final product



Design decisions

No agent can approve

API keys and MCP tools can only check and read. Approval needs a Reviewer or Administrator, and the end-to-end test proves a creator cannot self-approve.

Honest degradation

Missing OCR or embeddings are reported on the capabilities page and in the UI. Retrieval falls back to BM25 and TF-IDF, and text is never invented.

Calibrated evidence

Meaning-only matches need cosine ≥ 0.62, set above the highest similarity of unrelated queries, so unrelated passages are never cited.

Transparent OCR fixes

Only fixes that match the brand vocabulary or a clear price context are applied, and each is listed with its reason.

Secure integrations

Webhooks are HMAC-signed, retried and logged. Production allows only public HTTPS hosts (SSRF guard).

Fits a free host

About 300 MB of RAM with OCR and the embedding model loaded. Both models are baked into the Docker image.

Full detail: `submission/ROUND3_TECHNICAL_DETAILS.md`, `API_DOCUMENTATION.md` (Round 3 endpoints) and `DEPLOYMENT.md` §9.

05 · TESTING NOTES

Measured, including what did not work at first

Automated tests — 223

AREA	TESTS
Round 2 suites (integration, engines, security, governance, translation, unit, visual, config)	174
Improvement 1 — creative inspection	11
Improvement 2 — integrations	13
Improvement 3 — retrieval & ingestion	14
Foundation, demo seed	10
End-to-end workflow	1
Total — all pass on SQLite and PostgreSQL	**223**

The same 223 tests also pass on PostgreSQL 18 with pgvector (`BV_TEST_DATABASE_URL`), where the pgvector test runs too.

Evaluations

EVALUATION	RESULT
Creatives, main set (26)	F1 1.00 · OCR 99.7%
Creatives, held-out (12)	F1 1.00 · OCR 99.5%
Creatives, first run before OCR fixes	recall 0.86
Retrieval R@3, Round 2 → Round 3	0.675 → 0.825
Retrieval MRR	0.575 → 0.717
Paraphrase queries R@3	0.581 → 0.774
Exact-term queries R@3	1.000 (unchanged)
Text analyzer (36 cases)	P 1.00 · R 0.86 · F1 0.925

Honest reading.

- **Creative sets:** these are generated, with clean text on plain backgrounds, so real photos will score lower. The held-out set was written after tuning and run once without changes.
- **Retrieval corpus:** written by the team.
- **Dense threshold:** calibrated on unrelated queries, not on the labelled ones.

Manual verification

- **Creative Inspection:** a banner uploaded through the real API had every issue boxed on the right line.
- **Publishing page:** channel and disclaimer results checked.
- **Settings:** engine status, API keys and webhooks.
- **Notification bell:** showed a real event.
- **MCP:** a real stdio session with the official MCP client listed 5 tools, ran a check without saving anything, and the gate blocked unapproved copy.
- **Builds:** the production frontend build passed (37 pages) and the type check is clean.

06 · CHANGES TO THE PLAN

What differs from the Round 2 promises

ROUND 2 PLAN	WHAT WE DID	WHY
Tesseract OCR with a vision-model fallback	RapidOCR on ONNX, no vision fallback	Installs with pip, so it runs the same locally and on Render without system packages. A vision model would add cost and latency for little gain on clean text.
Hosted embedding API	A local BGE-small model by default; any OpenAI-compatible API is optional	No paid key is needed, results are reproducible, and it fits the free host's memory.
Optional cross-encoder reranker	Not built	Memory budget on the free host. RRF fusion already gave +15 points of recall@3.
Live deployment on managed PostgreSQL with row-level security	PostgreSQL support built and tested, with deployment steps for Neon. RLS not added	Switching the live site needs the team's Neon connection string. Tenant isolation stays app-level and tested.
Multi-hop evidence across document versions	Document-level version history: retired documents are no longer cited and their facts are flagged	Simpler and explainable. Fact-level "Outdated" already comes from Product Truth's fact history.
Slack and email notifications	In-app notifications are always on; Slack and email are optional	Email needs an SMTP account; the app shows "not configured" instead of failing silently.
Recall above 0.86	Retrieval recall improved; the text analyzer stays at 0.86	The analyzer's misses are rule coverage (wording it has no rule for), not retrieval.

07 · KNOWN LIMITATIONS

Stated openly

LIMITATION	DETAIL
Live data persistence	The live site uses SQLite on Render's temporary disk until a Neon `DATABASE_URL` is added. Demo data is re-seeded on every restart.
Creative accuracy	Measured on generated creatives. Photos, busy backgrounds and stylised fonts will be harder.
Video	6 sampled frames; text shown only between samples can be missed.
Retrieval	7 of 40 paraphrase queries are still missed (e.g. "release date"). No reranker.
Row-level security	Not added. Isolation is enforced by the application on every query and tested.
Email	Needs SMTP settings.
Free hosting	Cold starts of up to a minute; 512 MB of memory (about 300 MB used).
Thin real history	Reviewer and trend insights need more real use. The demo data is labelled.

08 · DEMO SCRIPT

Five minutes with the final product

TIME	STEP	WHAT TO SHOW
0:00	**Command Center**	Real KPIs and the demo campaigns.
0:30	**Creative Inspection**	Upload a banner with a wrong price: it is boxed, explained and corrected.
1:30	**Brand DNA**	Upload a Word guideline marked as replacing the old one. The Knowledge Gap Detector shows the change.
2:10	**Content Intelligence**	Paraphrased copy now cites the right guideline section.
2:50	**Approval → Passport**	A reviewer approves the corrected copy and a Brand Passport is issued.
3:30	**Publishing gate**	Settings → API key, then run the curl example: approved copy is allowed, edited copy is blocked, and the bell shows the event.
4:20	**MCP**	Ask an AI assistant to "check this copy with BrandVault". It reads but cannot approve.

09 · TEAM CONTRIBUTIONS

Who did what

> To be completed by the team. The repository history cannot show individual contributions because all commits were made from one GitHub account. Please fill in each member's real work; this report does not guess.

MEMBER	ROLE	CONTRIBUTIONS (ROUNDS 1–3)
Deepak Mouriya M (Team Leader)	Product & Frontend	*to be filled by the team*
Roshan Vikram AV	AI / Backend / Full-Stack	*to be filled by the team*
Pritheesh Kumar V	MCP, Data & RAG	*to be filled by the team*
Aishwarya R M	Visual DNA & QA	*to be filled by the team*

Parts of the implementation, tests and documentation were produced with an AI coding assistant working under the team's direction.

Conclusion

BrandVault AI now governs every form a campaign takes: typed copy, banners, decks, videos and translations. It finds evidence by meaning, and it checks content where teams write and publish it. Across all of it the principle from Round 1 holds: agents advise, evidence grounds, and humans decide.